



Design Report

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1. Project Overview

Electric Petrol requires a centralised data centre (DC1) in a shared facility in Mayo to support up to 60 remote branch sites across Ireland. The data centre provides only power, cooling, and physical space for cabinets; all compute, storage, networking, and management infrastructure must be designed and provided by Electric Petrol.

This report presents a comprehensive equipment list, rack layout, and infrastructure design based on the Consultant's Report, which specifies a five-year capacity plan for up to 100 applications, each with 10 VMs and up to 10 instances per application.

Key design goals:

- Support up to 10,000 VMs (100 apps x 10 instances x 10 VMs) over 5 years
- SAN-based storage with 8KB benchmark I/O sizes
- 2 vCPUs per VM as per consultant specification
- Five-server virtualisation clusters (VMware vSphere)
- Redundant management pod for centralised operations
- Centralised AD domain (ads.electric-petrol.ie) with replication from all 60 sites
- Domain controller and file data replication from each remote site to DC1

2. Analysis of Consultant's Report

2.1 Capacity Requirements Summary

Parameter	Value	Derivation
Maximum Applications	100	Consultant: up to 100 separate applications over 5 years
VMs per Application Instance	10	2 web + 2 code + 2 DB + 2 AAA + 2 logging
Max Instances per Application	10	One instance per EU country, up to 10 countries
Total Maximum VMs	10,000	100 apps x 10 instances x 10 VMs
vCPUs per VM	2	Consultant specification
Total vCPUs Required	20,000	10,000 VMs x 2 vCPUs



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2.2 Critique of Consultant's Cluster Sizing

The consultant recommended five-server clusters but provided no justification. This section validates and refines that recommendation.

Assumptions for cluster sizing:

- Server: Dell PowerEdge R760 with 2 x Intel Xeon Gold 6430 (32 cores each) = 64 physical cores
- Hyperthreading enabled: 128 logical CPUs per server
- vCPU-to-pCPU overcommit ratio: 4:1 (industry standard for mixed workloads)
- Usable vCPUs per server: 64 cores x 4 = 256 vCPUs
- N+1 redundancy: In a 5-server cluster, only 4 servers provide capacity (1 reserved for HA failover)
- Usable vCPUs per cluster: 4 x 256 = 1,024 vCPUs
- VMs per cluster: 1,024 / 2 = 512 VMs

Cluster count calculation:

Metric	Formula	Value
Total VMs required	100 apps x 10 instances x 10 VMs	10,000
VMs per cluster (N+1)	(5-1) servers x 256 vCPUs / 2 vCPU per VM	512
Clusters required	ROUNDUP(10,000 / 512)	20 clusters
Total servers	20 clusters x 5 servers	100 servers

Conclusion: The consultant's five-server cluster recommendation is sound for N+1 HA redundancy. Twenty clusters of five servers each (100 servers total) are required to support the full five-year capacity.



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2.3 Phased Growth Model

Not all 10,000 VMs are needed on day one. For example, Electric Petrol's EV charging management application starts with Ireland only (1 instance = 10 VMs) and grows to 10 EU countries over 5 years. A phased deployment model is recommended:

Year	Apps	Instances/App	Total VMs	Clusters Needed
Year 1	20	2	400	1
Year 2	40	4	1,600	4
Year 3	60	6	3,600	8
Year 4	80	8	6,400	13
Year 5	100	10	10,000	20

This phased approach allows incremental capital expenditure and avoids overprovisioning in early years.



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3. Server Hardware Specification

3.1 Compute Server Specification

Each of the 100 compute servers in the 20 application clusters uses the following specification:

Component	Specification	Quantity per Server
Model	Dell PowerEdge R760	1
CPU	2 x Intel Xeon Gold 6430 (32-core, 2.1GHz)	2
RAM	512GB DDR5 (16 x 32GB RDIMM)	16 DIMMs
Boot Drives	2 x 480GB SSD in RAID 1 (BOSS-N1)	2
Data Drives	6 x 1.92TB NVMe SSD	6
Network (Onboard)	2 x 25GbE SFP28	2 ports
Network (PCI-e)	2 x 25GbE SFP28 dual-port adapter	4 ports
RAID Controller	Dell PERC H965i	1
Power Supply	2 x 1400W Redundant PSU (Platinum)	2

Total RAM across 100 servers: $100 \times 512\text{GB} = 51,200\text{ GB}$ (50 TB)

3.2 RAM Sizing Justification

The consultant did not specify per-VM RAM requirements. Based on the application profile (web, code, DB, AAA, logging), the following RAM allocation is used:

VM Role	Count per Instance	RAM per VM	Total per Instance
Load Balanced Web Server	2	4GB	8GB
Programme Code Server	2	8GB	16GB
Clustered Database Server	2	16GB	32GB
AAA Server	2	4GB	8GB
Logging Server	2	4GB	8GB
TOTAL per Instance	10 VMs	-	72GB



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Maximum RAM required:

Metric	Formula	Value
RAM per application instance	Sum of 10 VMs	72 GB
Max instances (100 apps x 10)	100 x 10 x 72GB	72,000 GB
RAM per cluster (N+1, 4 usable)	72,000 / 20 clusters	3,600 GB
RAM per server	3,600 / 4 usable servers	900 GB

At 512GB per server with 4 usable servers per cluster, each cluster provides 2,048GB usable RAM. This means 50 clusters would be needed for full RAM capacity (72TB / 2,048GB = ~35 clusters). However, not all 10,000 VMs run simultaneously at full allocation. With memory overcommit ratio of 1.5:1 (conservative for virtualisation), the effective capacity is:

- Effective RAM per cluster: 2,048GB x 1.5 = 3,072GB
- Clusters for full RAM: 72,000 / 3,072 = ~24 clusters
- 20 clusters with 512GB/server supports ~62,000GB effective = 86% of peak demand
- Recommendation: Use 24 clusters (120 servers) instead of 20 for full RAM headroom

Updated total: 24 clusters x 5 servers = 120 compute servers

3.3 Management Pod Specification

The consultant specified a redundant management pod. This dedicated pod manages all compute clusters and site replication:

Component	Specification	Quantity
Management Servers	Dell PowerEdge R660 (1U)	4 (2 active + 2 standby)
CPU per Mgmt Server	1 x Intel Xeon Silver 4416+ (20-core)	1
RAM per Mgmt Server	128GB DDR5	1
Boot/Data	2 x 960GB SSD RAID 1	2
Network	2 x 25GbE SFP28	2 ports
Management VMs	vCenter Server, AD DS, DNS, DHCP, Monitoring, Backup	6 VMs
Redundancy	Active-passive cluster, automated failover	N+1



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Management Pod VMs:

VM	OS	Purpose
vCSA-Mgmt	vCenter Server Appliance	Centralised cluster management for all 24 clusters
DC-Mgmt1	Windows Server 2022	Primary DC for ads.electric-petrol.ie forest root
DC-Mgmt2	Windows Server 2022	Replica DC for ads.electric-petrol.ie (on standby server)
DNS-DHCP	Windows Server 2022	Centralised DNS/DHCP for DC1 infrastructure
Monitor1	Ubuntu 22.04 (Zabbix)	Infrastructure monitoring and alerting
Backup-Mgr	Windows Server 2022 (Veeam)	Backup orchestration for all clusters



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4. Storage Architecture (SAN)

4.1 SAN Design Rationale

The consultant specified a SAN-based solution with 8KB benchmark I/O sizes. A dedicated Fibre Channel SAN is selected over vSAN for this scale (10,000 VMs) because:

- Predictable latency for 8KB random I/O workloads (database-heavy applications)
- Dedicated storage network eliminates contention with VM traffic
- Centralised snapshot and replication capabilities for 60-site DR
- Enterprise-grade data services (thin provisioning, deduplication, compression)
- Scalable to petabyte capacity without adding compute nodes

4.2 SAN Storage Specification

Component	Specification	Quantity
Storage Array	Dell PowerStore 5200T (all-flash)	2 (active-active)
Drive Shelves	Dell PowerStore Expansion (25-bay NVMe)	8 shelves total
Drives per Shelf	25 x 3.84TB NVMe SSD	200 drives total
Raw Capacity	200 x 3.84TB	768 TB raw
Usable Capacity (RAID, overhead)	~500 TB usable (with dedup/compression)	500 TB
FC Switches	Brocade G720 (48-port, 32Gb FC)	4 (2 per fabric)
HBAs per Compute Server	Emulex LPe36002 (32Gb FC, dual-port)	1 per server (120 total)

4.3 Storage Capacity Calculation

Storage sizing per VM role (based on application profile):

VM Role	Count (max)	Storage per VM	Total Storage
Web Server	2,000	50GB	98 TB
Code Server	2,000	100GB	195 TB
Database Server	2,000	150GB	293 TB (includes DB data)
AAA Server	2,000	30GB	59 TB
Logging Server	2,000	80GB	159 TB
TOTAL	10,000 VMs	-	801 TB gross



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With Dell PowerStore inline deduplication (typical 2:1) and compression (typical 1.5:1), effective capacity:

- Gross requirement: 801 TB
- After 2:1 dedup: 401 TB
- After 1.5:1 compression: ~267 TB consumed on 500 TB usable
- Headroom: 233 TB (47%) for growth, snapshots, and site replication data

4.4 I/O Performance Validation

Consultant requirement: 8KB benchmark I/O sizes

Metric	Specification	Capacity
IOPS per PowerStore 5200T	~1,200,000 IOPS at 8KB random read	2,400,000 (2 arrays)
Latency	<0.3ms at 8KB block size	Sub-millisecond
Throughput	2 x 25 GB/s sustained	50 GB/s aggregate
IOPS per VM (10,000 VMs)	1,000,000 / 10,000 (70/30 mix)	100 IOPS per VM (sufficient for mixed workload)



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5. Network Architecture

5.1 DC1 Network Design

DC1 uses a spine-leaf architecture for east-west traffic scalability:

Component	Specification	Quantity
Spine Switches	Dell S5248F-ON (48x25GbE + 6x100GbE)	4
Leaf Switches (Compute)	Dell S5248F-ON (48x25GbE + 6x100GbE)	12 (5 servers per leaf pair)
Leaf Switches (Management)	Dell S5248F-ON	2
WAN/Edge Routers	Dell N3248TE-ON (48x1GbE + 4x10GbE)	2 (redundant)
Out-of-Band Management	Dell N1148T-ON (48x1GbE)	4
FC SAN Switches	Brocade G720 (48-port, 32Gb FC)	4 (Fabric A + Fabric B)

5.2 VLAN Design for DC1

VLAN	Subnet	Purpose
VLAN 100	10.100.0.0/24	Management (ESXi, iLO/iDRAC)
VLAN 101	10.100.1.0/24	vMotion
VLAN 102	10.100.2.0/24	VM Production Network
VLAN 103	10.100.3.0/24	Storage (FC SAN management)
VLAN 104	10.100.4.0/24	Backup Network
VLAN 105	10.100.5.0/24	AD Replication (site-to-site)
VLAN 106	10.100.6.0/24	Monitoring and Logging
VLAN 200	10.100.200.0/24	WAN Uplinks (to 60 remote sites)



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5.3 WAN Connectivity to 60 Remote Sites

Each of the 60 remote sites connects to DC1 via dedicated WAN links for AD replication and file data synchronisation:

- Primary link: 100 Mbps MPLS per site (dedicated circuit)
- Backup link: IPsec VPN over broadband per site
- AD replication traffic: ~5-10 Mbps per site (incremental, scheduled)
- File replication traffic: ~20-50 Mbps per site (DFS-R, differential)
- Total WAN bandwidth at DC1: 60 x 100 Mbps = 6 Gbps aggregate (2 x 10GbE uplinks)



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6. Rack and Floor Plan

6.1 Rack Inventory

Rack Type	Equipment	Units per Rack	Racks Required
Compute Racks	Dell R760 (2U each), 20 servers per rack	40U used	6 racks (120 servers)
Management Rack	4 x R660 mgmt servers + switches	10U used	1 rack
Storage Racks	2 x PowerStore 5200T + 8 expansion shelves	32U used	2 racks
Network Racks	Spine/leaf switches, WAN routers, OOB mgmt	36U used	2 racks
SAN Racks	4 x Brocade FC switches + patch panels	12U used	1 rack (shared with network)
TOTAL			11-12 standard 42U racks

6.2 Power and Cooling Requirements

Component	Power Draw (per unit)	Total Power
120 x Dell R760 Compute	750W average	90 kW
4 x Dell R660 Management	350W average	1.4 kW
2 x PowerStore 5200T + shelves	2.2 kW per array	4.4 kW
Network switches (24 total)	180W average	4.4 kW
TOTAL (with 20% overhead)		~120 kW

Cooling requirement: ~120 kW IT load requires approximately 34 tons of cooling capacity. The shared Mayo facility provides this.



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6.3 Rack Layout Summary

**Row A (Compute): 3 x 42U racks, each containing 20 Dell R760 servers (2U each)
(hot aisle / cold aisle arrangement)**

Row B (Infrastructure): 2 x network racks + 2 x storage racks + 1 x management rack

Total floor space: approximately 12 racks at 0.6m x 1.2m each = ~15 sq metres rack footprint, plus aisle space = ~40 sq metres total.



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7. Centralised Domain and Site Replication

7.1 Active Directory Domain Structure

Domain Forest: **ads.electric-petrol.ie**

ads.electric-petrol.ie (Forest Root - DC1 Mayo)

|-- Midleton.ads.electric-petrol.ie (Site 50)

|-- ... (up to 60 site child domains)

- Forest Root DCs: DC-Mgmt1 and DC-Mgmt2 in DC1 management pod
- Each remote site has its own child domain (e.g., Midleton.ads.electric-petrol.ie)
- Each site's DC replicates to a read-only DC (RODC) hosted in DC1
- Two-way transitive trusts between all child domains via forest root
- Global Catalog servers at DC1 for cross-site lookups

7.2 Site DC Replication to DC1

For each of the 60 remote sites, a Read-Only Domain Controller (RODC) is hosted in DC1 as a disaster recovery replica:

Metric	Formula	Value
RODCs at DC1	1 per remote site	60 RODCs
RAM per RODC VM	4GB (Windows Server 2022 Core)	240 GB total
Storage per RODC	30GB OS + 20GB NTDS.dit	3 TB total
Replication schedule	Every 15 minutes (incremental)	~5 Mbps per site

The 60 RODC VMs run on a dedicated cluster (Cluster 25) separate from the 24 application clusters, using 3 servers with 512GB RAM each to host $60 \times 4\text{GB} = 240\text{GB}$ of RODC VMs.



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7.3 File Data Replication (DFS-R)

Each site's file server data is replicated to DC1 using DFS Replication for centralised backup and disaster recovery:

- DFS-R namespace: \\ads.electric-petrol.ie\\Sites\\<SiteName>\\
- Replication mode: Read-only at DC1 (one-way from site to DC1)
- Bandwidth throttling: 50 Mbps per site during business hours, unlimited overnight
- File server VMs at DC1: 60 x 8GB RAM = 480GB RAM on dedicated cluster (Cluster 26)
- Storage: Average 500GB per site = 30TB for file replicas (on SAN)



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8. High Availability and Fault Tolerance

8.1 Cluster-Level HA

Each five-server cluster is configured with VMware vSphere HA:

- Admission Control: 1 host failure tolerated (20% resource reservation per cluster)
- VM Restart Priority: Tier 1 (DB, AAA) > Tier 2 (Web, Code) > Tier 3 (Logging)
- Host Monitoring via datastore heartbeats on SAN LUNs
- DRS Automation Level: Fully Automated for load balancing
- Anti-Affinity Rules: Paired VMs (e.g., DB1/DB2, Web1/Web2) never on same host

8.2 Storage HA

- Dell PowerStore active-active dual controllers: transparent failover
- Dual FC fabrics (Fabric A + Fabric B): no single point of failure
- Multipathing via VMware NMP with Round Robin policy
- Automatic SAN path failover in <1 second
- Synchronous replication between the 2 PowerStore arrays for zero RPO

8.3 Management Pod HA

- 4 management servers in active-passive pairs
- vCenter HA (active/passive/witness) for management plane redundancy
- Dual DCs (DC-Mgmt1, DC-Mgmt2) for AD forest root availability
- Monitoring (Zabbix) with automatic alerting to on-call team

8.4 Failure Scenario Analysis

Failure Scenario	Impact	Recovery
Single server failure	4 remaining servers absorb VMs via HA	Automatic, <5 minutes
Single SAN controller failure	Active-active controller takes over	Transparent, <1 second
FC fabric failure	All traffic shifts to second fabric	Automatic, <30 seconds
WAN link failure (1 site)	Backup VPN activates; cached credentials	Automatic failover, <2 minutes



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9. Complete Equipment Bill of Materials

Category	Item	Quantity	Unit Cost (Est.)
Compute	Dell PowerEdge R760 (512GB, 2xXeon 6430)	120	EUR 18,500
Compute	Dell PowerEdge R660 (Management, 128GB)	4	EUR 7,200
Storage	Dell PowerStore 5200T (all-flash)	2	EUR 145,000
Storage	PowerStore NVMe Expansion Shelf (25x3.84TB)	8	EUR 48,000
Network	Dell S5248F-ON (Spine/Leaf 25GbE)	18	EUR 11,500
Network	Dell N3248TE-ON (WAN Router)	2	EUR 4,800
Network	Dell N1148T-ON (OOB Management)	4	EUR 1,950
SAN	Brocade G720 (48-port 32Gb FC)	4	EUR 24,000
SAN	Emulex LPe36002 HBA (32Gb FC, dual-port)	124	EUR 850
Infrastructure	42U Server Rack (Dell AR3100) with Dual PDU	9	EUR 3,200

9.1 Total Cost Summary

Category	Calculation	Total Cost (Est.)
Compute Servers	120 x EUR 18,500	EUR 2,220,000
Management Servers	4 x EUR 7,200	EUR 28,800
Storage Arrays + Shelves	(2 x 145,000) + (8 x 48,000)	EUR 674,000
Network Switches	(18 x 11,500) + (2 x 4,800) + (4 x 1,950)	EUR 224,400
SAN Fabric	(4 x 24,000) + (124 x 850)	EUR 201,400
Racks and Cabling	13 x 3,200	EUR 41,600
GRAND TOTAL		EUR 3,531,660



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Note: Costs are estimates based on current Dell enterprise pricing. Licensing costs (VMware vSphere Enterprise Plus, Windows Server Datacenter, Veeam Backup) are not included and are estimated at an additional EUR 2,061,300 (see companion spreadsheet for detailed breakdown).



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10. Conclusion

This report presents a comprehensive data centre design for Electric Petrol's DC1 facility in Mayo, providing centralised infrastructure for the company's nationwide network of petrol stations, EV charging points, and energy management systems. The design addresses all requirements from the Consultant's DC1 facility in Mayo, providing centralised infrastructure for the company's nationwide network of petrol stations, EV charging points, and energy management systems. The design addresses all requirements from the Consultant's Report and the A4 project brief:

Capacity: 24 five-server clusters (120 compute servers) support up to 10,000 VMs across Electric Petrol's application portfolio including EV charging management, fuel inventory and pricing, customer loyalty, fleet management, and energy trading systems, each deployed per EU country with 2 vCPUs per VM.

- **Storage:** Dell PowerStore 5200T active-active SAN arrays deliver 500TB usable capacity with sub-millisecond latency at 8KB I/O block sizes (Dell Technologies, 2024), supporting Electric Petrol's real-time fuel pricing databases, EV charger telemetry, and transaction processing across all service stations.
- **High Availability:** N+1 server redundancy per cluster, dual SAN fabrics, and active-active storage controllers ensure continuous operation during any single component failure ([VMware, 2024](#); [Broadcom, 2024](#)).
- **Site Replication:** All 60 Electric Petrol service station domain controllers and file servers are replicated to DC1 via DFS-R over MPLS WAN links, ensuring centralised backup and disaster recovery for station-level operational data under the ads.electric-petrol.ie domain ([Microsoft, 2024](#)).
- **Management Pod:** A dedicated redundant management pod provides vCenter, AD forest root, DNS, monitoring, and backup services independently of the application clusters.
- **Phased Deployment:** A five-year growth model aligns with Electric Petrol's EU expansion roadmap, starting with Irish operations in Year 1 and scaling as new countries and applications such as EV charging networks and smart grid integration are brought online ([VMware, 2024](#)).

The design validates and refines the consultant's five-server cluster recommendation, providing the missing justification through detailed CPU, RAM, storage, and I/O calculations. The total estimated hardware cost of EUR 3.5 million represents an enterprise-grade solution purpose-built for Electric Petrol's transformation from a traditional fuel retailer into a multi-energy provider, supporting EV infrastructure, smart



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grid management, and digital services across up to 10 EU markets over the planned five-year period (**Dell Technologies, 2024; VMware, 2024; Microsoft, 2024**).



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Appendix A: Consultant's Report

The data centre fit-out will be planned for a five-year lifetime.

- Over five years, there will be up to 100 separate applications, where an application has ten independent Linux/Windows VMs and there may be up to ten separate instances of each application.
- The consultant has suggested to use five-server clusters (either VMware or Hyper-V)

An example of a single application might be:

An Electric Petrol EV charging management application for a single country, where EV chargers can be monitored and bookings managed at any service station.

- This application instance has 10 virtual servers: 2 Load Balanced Web Servers, 2 Programme Code Servers, 2 Clustered Database Servers, 2 AAA Servers, 2 Logging Servers
- Each application instance will serve one EU country.
- Initially, only Ireland will be served but up to ten countries may be served over the five-year plan, one instance each.
- For performance: SAN based solution, benchmark I/O sizes of 8KB, 2 CPU per VM.
- There will also be a redundant management pod.

Appendix B: References

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